Day 107 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Find Parallelogram and Trapezoid Area

Standards: 6.G.A.1

Learning target: I can find parallelogram and trapezoid areas by rearranging shapes.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Cut and move a corner of a parallelogram to make a rectangle: A = bh. Two matching trapezoids form a parallelogram, giving A = (b1 + b2)h/2. Heights must be perpendicular.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 23

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01 / Learn

Cut and move a corner of a parallelogram to make a rectangle: A = bh. Two matching trapezoids form a parallelogram, giving A = (b1 + b2)h/2. Heights must be perpendicular.

Worked example 1

A parallelogram has base 2 m and perpendicular height 5 m. Sketch how it can be rearranged into a rectangle and find its area.

1. Cut and move a corner of a parallelogram to make a rectangle: A = bh.

2. Two matching trapezoids form a parallelogram, giving A = (b1 + b2)h/2.

3. Cut and shift a corner

4. A = bh = 2 × 5 = 10.

5. State the result in the form requested: 10 m².

Answer: 10 m²

Worked example 2

A trapezoid has parallel bases 14 m and 22 m, and perpendicular height 3 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

1. Cut and move a corner of a parallelogram to make a rectangle: A = bh.

2. Use average base × height: (14 + 22)/2 × 3 = 54.

3. A full rectangle at the longer base includes extra corner regions.

Answer: 54 m²

Day 107 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A parallelogram has base 8 m and perpendicular height 4 m. Sketch how it can be rearranged into a rectangle and find its area.

02 / Constructed response

A parallelogram has base 6 m and perpendicular height 4 m. Sketch how it can be rearranged into a rectangle and find its area.

03 / Constructed response

A parallelogram has base 8 m and perpendicular height 7 m. Sketch how it can be rearranged into a rectangle and find its area.

04 / Constructed response

A trapezoid has parallel bases 4 m and 14 m and perpendicular height 5 m. Sketch it and find its area.

05 / Constructed response

A trapezoid has parallel bases 8 m and 16 m and perpendicular height 7 m. Sketch it and find its area.

06 / Constructed response

A trapezoid has parallel bases 8 m and 12 m and perpendicular height 4 m. Sketch it and find its area.

07 / Constructed response

A trapezoid has parallel bases 6 m and 14 m and perpendicular height 5 m. Sketch it and find its area.

08 / Numeric entry and error analysis

A trapezoid has parallel bases 14 m and 22 m, and perpendicular height 4 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

09 / Numeric entry and error analysis

A trapezoid has parallel bases 4 m and 8 m, and perpendicular height 2 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

10 / Constructed response

A trapezoid has parallel bases 12 m and 24 m and perpendicular height 2 m. Sketch it and find its area.

Day 107 | Unit 08 | Answer key

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1. 32 m²

Cut and shift a corner; A = bh = 8 × 4 = 32.

6.G.A.1 | Constructed response

2. 24 m²

Cut and shift a corner; A = bh = 6 × 4 = 24.

6.G.A.1 | Constructed response

3. 56 m²

Cut and shift a corner; A = bh = 8 × 7 = 56.

6.G.A.1 | Constructed response

4. 45 m²

A = (4 + 14) × 5 ÷ 2 = 45.

6.G.A.1 | Constructed response

5. 84 m²

A = (8 + 16) × 7 ÷ 2 = 84.

6.G.A.1 | Constructed response

6. 40 m²

A = (8 + 12) × 4 ÷ 2 = 40.

6.G.A.1 | Constructed response

7. 50 m²

A = (6 + 14) × 5 ÷ 2 = 50.

6.G.A.1 | Constructed response

8. 72 m²

Use average base × height: (14 + 22)/2 × 4 = 72. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis

9. 12 m²

Use average base × height: (4 + 8)/2 × 2 = 12. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis

10. 36 m²

A = (12 + 24) × 2 ÷ 2 = 36.

6.G.A.1 | Constructed response